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Artificial intelligence for binary dental caries diagnosis using intraoral images and dental radiographs: a systematic review and meta-analysis

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Abstract

Objectives Artificial intelligence (AI) has shown increasing potential in dental diagnostics, yet its accuracy for binary classification of dental caries across different imaging modalities remains unclear. This study aimed to systematically evaluate the diagnostic performance of AI models using clinical intraoral images and dental radiographs.

Methods Following the PRISMA-DTA guidelines, PubMed, Embase, Scopus, Web of Science, and IEEE Xplore were systematically searched for studies published between January 2015 and June 2025. Eligible studies applied AI models for caries diagnosis with extractable sensitivity and specificity. Data on dentition, dataset, analysis unit, caries prevalence in test dataset, and preprocessing methods were extracted. Reporting quality and risk of bias were assessed using CLAIM and QUADAS-2. Pooled estimates were calculated with a bivariate random-effects model, with subgroup analyses by image type and analytical unit.

Results 25 studies met the criteria, and 13 were included in the meta-analysis. Pooled sensitivity, specificity, and the area under the curve (AUC) were 0.86, 0.91 and 0.94, respectively. Intraoral image–based models achieved higher sensitivity (0.88) and AUC (0.95), while radiograph-based models showed higher specificity (0.92). Tooth-level analyses yielded stable, clinically relevant performance (0.87/0.91). High heterogeneity ($I^2 > 90\%$) was partly explained by image type, model architecture, reference standard variation, and test set caries prevalence.

Conclusion AI models showed good diagnostic accuracy for caries detection across imaging modalities and analytical units. However, given the substantial heterogeneity and limitations in study quality and reference standards, these summary estimates should be interpreted with caution. AI-based systems may serve as complementary decision-support tools in clinical practice, but further standardization, external validation, and high-quality multicenter studies are required before broad clinical implementation.

Keywords Artificial intelligence, Deep learning, Dental caries, Diagnostic accuracy, Systematic review, Meta-analysis

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Introduction

Dental caries is one of the most prevalent chronic diseases worldwide and affects individuals across all age groups [1]. Advanced carious lesions may progress to pulpitis and apical periodontitis, ultimately leading to tooth loss and adverse systemic consequences. In addition to its clinical impact, dental caries imposes a substantial economic burden on both individuals and healthcare systems. According to the World Health Organization, the direct and indirect costs associated with the treatment of dental caries amount to billions of U.S. dollars annually, representing a major public health challenge [2–4]. Early detection and accurate diagnosis are therefore critical for preventing disease progression and enabling timely, minimally invasive intervention. However, conventional diagnostic approaches—such as visual inspection, tactile probing, and radiographic interpretation—are inherently subjective and highly dependent on clinician experience, particularly for early-stage and proximal lesions. These limitations are further amplified in regions with limited access to trained dental professionals, underscoring the need for scalable, objective, and reliable diagnostic tools [5].

In recent years, artificial intelligence (AI), particularly deep learning–based approaches, has emerged as a powerful tool for image-based medical diagnosis. In dentistry, AI models have been applied to a wide range of dental imaging modalities, including intraoral photographs and various forms of radiography, to support caries detection [6–20]. Growing evidence suggests that AI systems can extract clinically relevant features from these images and achieve diagnostic performance comparable to, and in some settings exceeding, that of human examiners, while also offering advantages in speed and consistency [8, 21–23]. These developments have generated considerable interest in the potential role of AI as a decision-support tool for dental caries diagnosis.

Despite this rapidly expanding body of literature, substantial methodological heterogeneity exists across published studies. Variations in dataset composition, image type, diagnostic unit (e.g., tooth-level, surface-level, or image-level), reference standards, and model architecture complicate the interpretation and comparability of reported results [24]. Moreover, AI-based caries studies have encompassed a wide range of analytical tasks, including lesion segmentation, detection, and diagnostic classification, further contributing to inconsistencies in outcome reporting and performance metrics.

Two recent systematic reviews have evaluated the diagnostic accuracy of AI for dental caries detection using bitewing radiographs, with a particular focus on approximal or proximal lesions [25, 26]. While these studies provided important evidence for radiograph-based AI, their scope was restricted to a single imaging modality and did not examine how diagnostic performance varies across different image types, analytical units, or reporting quality. In addition, these reviews did not focus exclusively on binary diagnostic classification. In contrast, the present review was designed to synthesize evidence across multiple imaging modalities—including intraoral photographs, near-infrared and fluorescence-based imaging, as well as panoramic and periapical radiographs—while focusing specifically on binary caries diagnosis and integrating PRISMA-DTA, CLAIM, and QUADAS-2 to jointly evaluate diagnostic performance, reporting quality, and risk of bias.

Therefore, the present systematic review and meta-analysis was designed to address these gaps by quantitatively evaluating the diagnostic accuracy of AI models for binary classification of dental caries (presence versus absence) using clinical imaging data. By integrating diagnostic performance with rigorous assessment of reporting quality, risk of bias, and heterogeneity, this study aims to provide a robust evidence base for the clinical translation of AI in caries diagnosis and to identify priorities for future research.

Methods

This systematic review was designed and reported in accordance with the PRISMA - Diagnostic Test Accuracy (PRISMA-DTA) Checklist to enhance transparency and reproducibility in diagnostic accuracy reviews [27]. The study protocol was prospectively registered at PROSPERO (CRD420251108203).

Eligibility criteria

Inclusion criteria

Studies were included if they met all of the following conditions: (i) employed AI models to perform binary diagnostic classification of dental caries using either dental radiographs or clinical intraoral images; (ii) used alternative classification schemes but reported data that could be converted into a binary diagnostic outcome; (iii) reported at least one standard diagnostic accuracy metric, such as sensitivity, specificity, accuracy, or the area under the receiver operating characteristic curve (AUC); (iv) were published between January 2015 and June 2025.

Exclusion criteria

Studies were excluded if they: (i) focused exclusively on image segmentation or lesion localization without performing binary diagnostic classification; (ii) did not report diagnostic performance metrics or lacked essential methodological details; (iii) were non-original research articles, such as reviews, conference abstracts, case reports, or commentaries.

Information sources and search

A systematic search was conducted in the following electronic databases: PubMed/MEDLINE, Embase, Web of Science, Scopus, Google Scholar, and IEEE Xplore. To ensure reproducibility and avoid uncontrolled retrieval, the Google Scholar search was restricted to the first 200 results sorted by relevance, and the same predefined search terms and inclusion criteria were applied. The search covered literature published between January 2015 and June 2025. The lower time limit was selected because modern deep learning–based methods for medical image analysis became established after 2015, whereas earlier studies primarily relied on traditional machine-learning techniques that are not directly comparable to contemporary AI models. Search strategies combined keywords and Medical Subject Headings (MeSH) related to dental caries and artificial intelligence, as detailed in Table 1. This strategy is consistent with recent systematic reviews in this field and was designed to capture the full spectrum of AI-based caries detection studies while minimizing irrelevant retrieval.

Study selection

All retrieved records were initially screened independently by two reviewers based on titles and abstracts, with duplicate entries removed. Studies meeting the inclusion criteria proceeded to full-text evaluation. Any disagreements during the selection process were resolved through discussion, and a third independent reviewer resolved any disagreements.

Data collection and extraction

Two reviewers independently extracted key variables using a predesigned, standardized data collection form. The extracted information included: author and year of publication, imaging modality, dentition, dataset size and split (training/validation/test), unit of analysis, caries prevalence in the test dataset, reference standard, inter-observer agreement when reported (e.g., kappa statistics), image preprocessing and augmentation methods, AI platform or model architecture, and diagnostic performance metrics.

Reporting standards assessment

To systematically evaluate reporting quality and methodological transparency, the Checklist for Artificial Intelligence in Medical Imaging (CLAIM) [28] was applied to all eligible studies. CLAIM is a standardized framework for assessing key aspects of AI-based medical imaging research, including model development, data handling, result reporting, and reproducibility. A predefined subset of eight CLAIM items most relevant to diagnostic accuracy studies was selected, and each item was evaluated according to the original definitions proposed by Mongan et al. These domains included: (i) model type and construction; (ii) data source and image acquisition; (iii) image preprocessing and augmentation; (iv) dataset partitioning; (v) performance metrics; (vi) model interpretability; (vii) external validation and generalizability; and (viii) code availability or reproducibility. Each domain was rated based on whether the corresponding information was clearly and explicitly reported, and overall reporting quality was categorized as high, moderate, or low.

Table 1 Search strategy

Search	Topic and terms
#1	Artificial Intelligence:"artificial intelligence" OR "deep learning" OR "machine learning" OR "neural network" OR "computer vision"
#2	Dentistry:"dental caries" OR "tooth decay" OR "carious lesion"
#3	Search #1 and #2

Risk of bias (RoB) assessment

The RoB for all included diagnostic studies was independently assessed by two reviewers using the Quality Assessment of Diagnostic Accuracy Studies-2 (QUADAS-2) tool [29]. Any disagreements were resolved through discussion, and adjudication was performed by a third reviewer when necessary. QUADAS-2 evaluated study quality across four key domains: patient selection, index test, reference standard, flow and timing. In addition, applicability concerns in this review were assessed based on the first three domains. Studies were marked as low RoB, high RoB, or unclear RoB.

Quantitative synthesis

Quantitative synthesis was performed using the MIDAS v3.0 module and the METANDI command in STATA version 17 (StataCorp, College Station, TX, USA) [30]. Only studies with complete or derivable 2 × 2 contingency tables were included in the meta-analysis. A bivariate random-effects model was applied to estimate the pooled sensitivity and specificity of AI models in the diagnosis of dental caries and to generate a summary receiver operating characteristic (SROC) curve. Between-study heterogeneity was assessed using the I^2 statistic and by visual inspection of forest plots, while publication bias was evaluated using Deeks' funnel plot asymmetry test.

Certainty of evidence

The certainty of evidence for pooled diagnostic estimates was assessed using the GRADE approach, considering risk of bias, inconsistency, indirectness, imprecision, and publication bias.

Results

Search and selection

A total of 1,538 records were initially retrieved. After removing duplicates, 1,243 records remained. Following title and abstract screening, 113 articles were assessed as potentially eligible. After full-text evaluation, 25 studies met the inclusion criteria for the qualitative synthesis, of which 13 were eligible for inclusion in the meta-analysis. The detailed selection process is illustrated in the PRISMA flow diagram (Fig. 1).

Basic characteristics of included studies

All 25 included studies [11, 12, 18, 19, 31–51] applied artificial intelligence models for dental caries detection based on dental imaging data. Their detailed characteristics are summarized in Supplementary Table S1, and a high-level overview is provided in Table 2. The included articles were published between 2018 and 2024, with the majority appearing after 2021 and a peak in 2024 ($n = 11$). Although the literature search extended to June 2025, no studies published in 2025 met the predefined inclusion criteria.

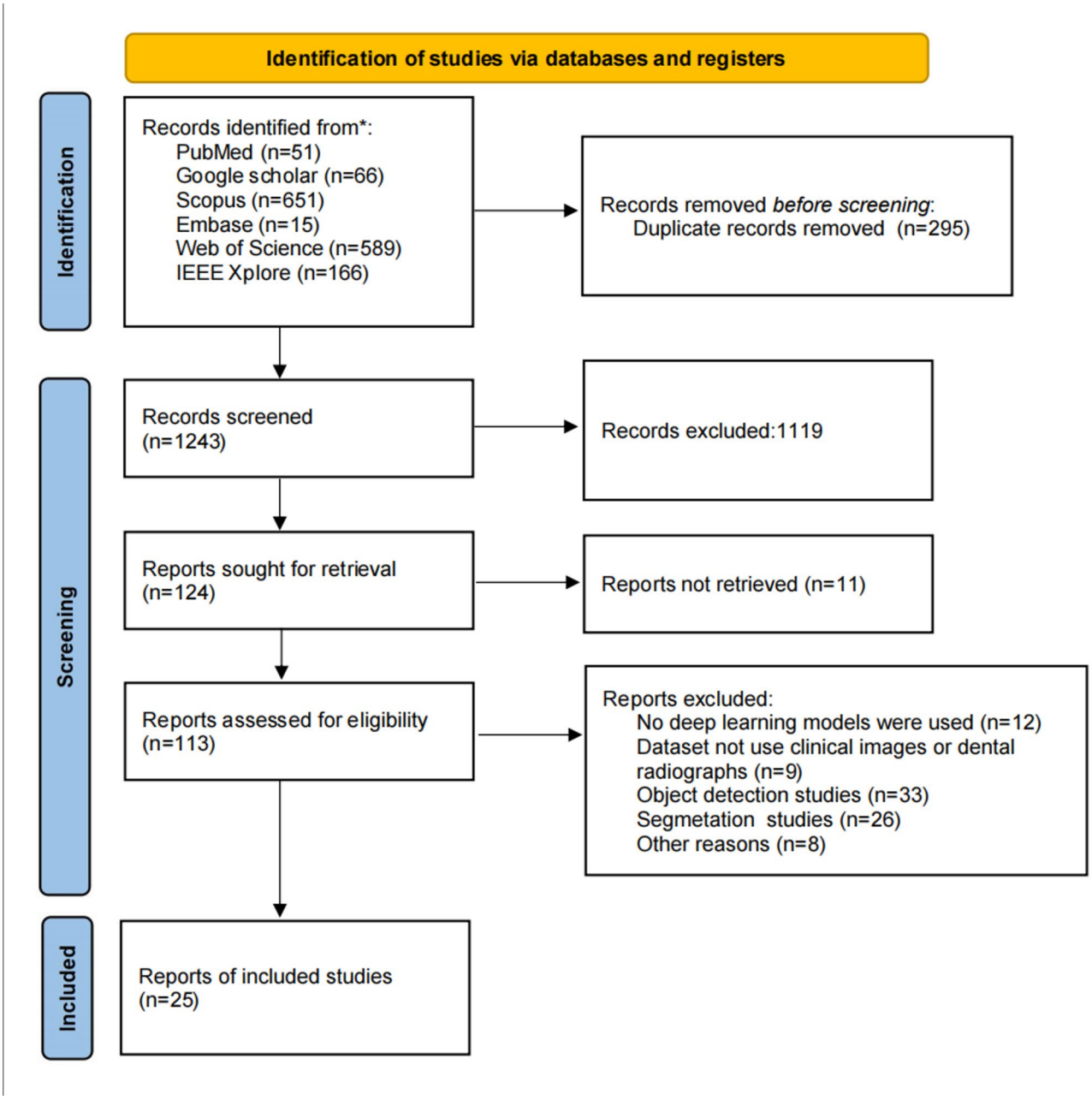


Fig. 1 PRISMA flow diagram illustrating the search strategy

Table 2 Overview of characteristics of included studies (n = 25)

Category	No. Of studies (%)
Imaging modality	
BWR	3 (12%)
PA	1 (4%)
PAN	7 (28%)
IOP/QLF/NILT	11 (44%)
Mixed modalities	3 (12%)
Unit of analysis	
Tooth-level	17 (68%)
Image-level	5 (20%)
Surface-level	1 (4%)
NR	3 (12%)
Dentition	
PER	11 (44%)
PRI	2 (8%)
Mixed	3 (12%)
NR	9 (36%)
Reference standard	
≥ 2 experts	13 (52%)
Single reader	7 (28%)
Diagnostic report	1 (4%)
Atificial lesions (no clinical reference)	1 (4%)
NR	3 (12%)
Inter-observer agreement reported	
κ reported	4 (16%)
NR	21 (84%)

BWR Bitewing radiographs, *PA* Periapical radiographs, *PAN* Panoramic radiographs, *IOP* Intraoral photographs, *QLF* Quantitative light-induced fluorescence, *NILT* Near-infrared light transillumination, *PER* Permanent dentition, *PRI* Primary dentition, *NR* Not reported, *κ* kappa statistic for inter-observer agreement

Reference standard categories include ≥2 experts (two or more human examiners, with or without formal consensus), single reader, clinical records/diagnostic report (medical or diagnostic records as ground truth), and artificial lesions (caries artificially generated rather than derived from clinical patient data). One study reported both tooth-level and image-level outcomes and was counted in both categories

Across the included studies, a wide range of imaging modalities, analytical units, and modeling approaches was observed (Supplementary Table S1), reflecting substantial methodological diversity in this field. Most studies relied on clinical radiographs or intraoral images, whereas only a small number used ex vivo specimens or artificially generated lesions. Caries prevalence in the test sets varied markedly, spanning from low-prevalence screening-type datasets to high-prevalence diagnostic cohorts.

The AI models evaluated encompassed conventional convolutional neural networks, transformer-based architectures, and self-supervised learning frameworks. Most studies assessed diagnostic performance at the tooth level, while fewer adopted image-level or surface-level units of analysis. Image preprocessing and data augmentation were commonly applied to enhance model robustness. Reference standards were typically based on expert annotation, although reporting of inter-observer agreement was limited, with kappa statistics available in only a small subset of studies.

CLAIM checklist

In the included AI-based diagnostic studies, methodological quality was evaluated using the eight core items of the CLAIM checklist. Most studies provided reasonably comprehensive descriptions of the model type, image source, image preprocessing procedures, and reported performance metrics. Nevertheless, several recurring limitations were observed. First, external validation was lacking in 70% of the studies, as no independent external test set was employed, thereby limiting the assessment of model generalizability. Second, model interpretability was insufficient in 62% of the studies, which failed to provide visual explanations (e.g., Grad-CAM) or to elucidate the underlying decision-making process. Third, code availability and reproducibility were restricted—only one study provided an online link to the model, whereas the remainder did not disclose source code or model specifications, hampering transparency and reproducibility. Overall, based on total scores, one study (4%) was classified as "high" quality, 21 (84%) as "moderate," and three (12%) as "low."

RoB assessment

Detailed assessments of risk of bias and applicability concerns are provided in Supplementary Table S2. RoB was assessed using the QUADAS-2 tool. 12 studies (48%) demonstrated a low RoB across all four domains, and 14 studies (56.0%) were judged to have a low risk in terms of applicability. Among the four domains, the highest proportion of high RoB was observed in the "reference standard" domain, where only 17 studies (68%) were classified as low risk. The main reasons included unclear definition of the reference standard, lack of a standardized criterion, or absence of blinded comparison.

Of the 25 included studies, 13 met the eligibility criteria for meta-analysis. The remaining 12 studies were excluded due to factors such as in vitro study design, use of artificially generated caries images, unclear reference standards, insufficient description of the AI model, or inability to extract a 2×2 contingency table.

Quantitative analysis

High between-study heterogeneity was observed, with I^2 values exceeding 90% for both sensitivity and specificity, indicating substantial variability across the included studies. Despite this, the pooled analysis demonstrated that AI models for caries diagnosis achieved a combined sensitivity of 0.86 (95% CI: 0.82–0.89) and a combined specificity of 0.91 (95% CI: 0.88–0.94) (Fig. 2). The SROC curve yielded an AUC of 0.94 (95% CI: 0.92–0.96),

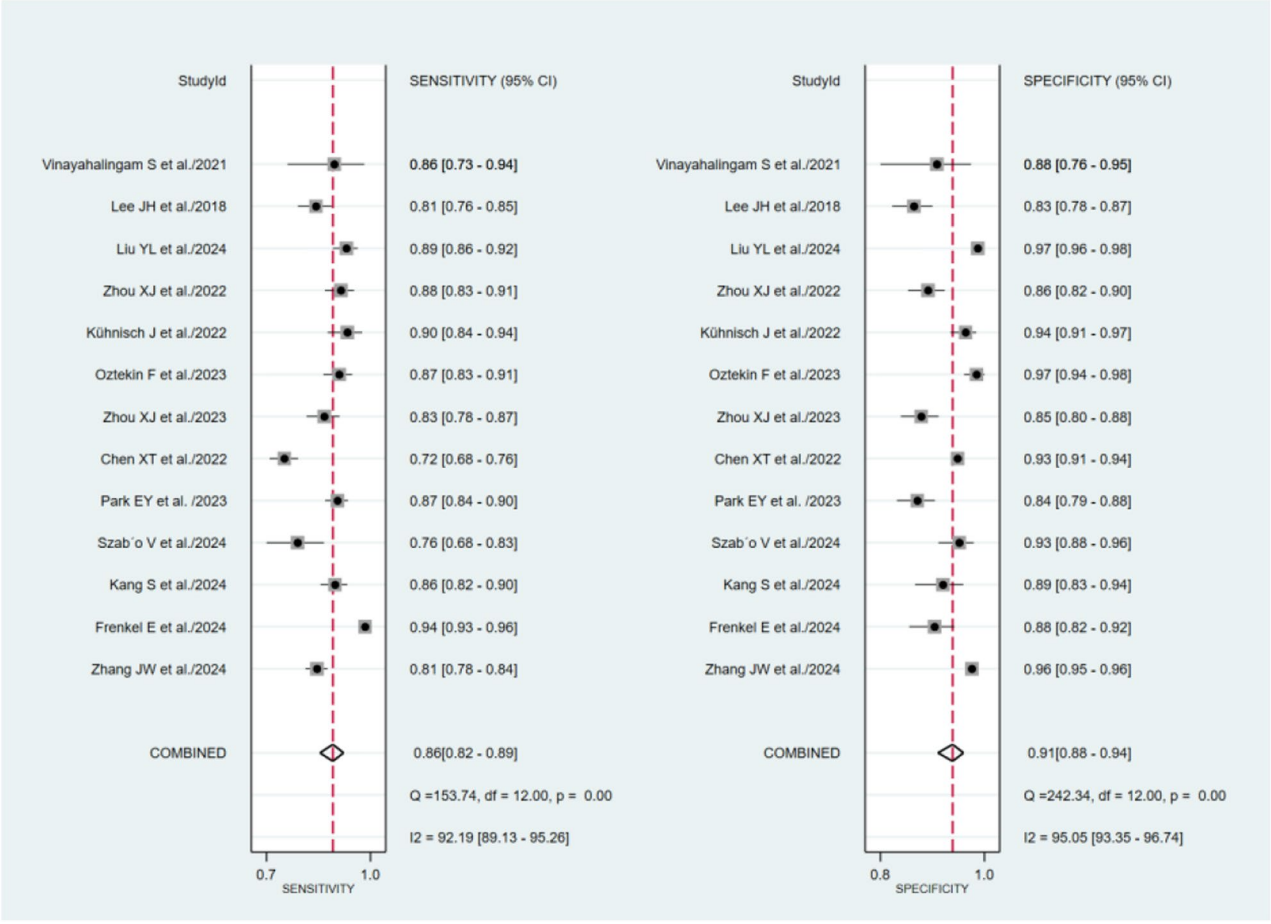


Fig. 2 Forest plot for sensitivity and specificity

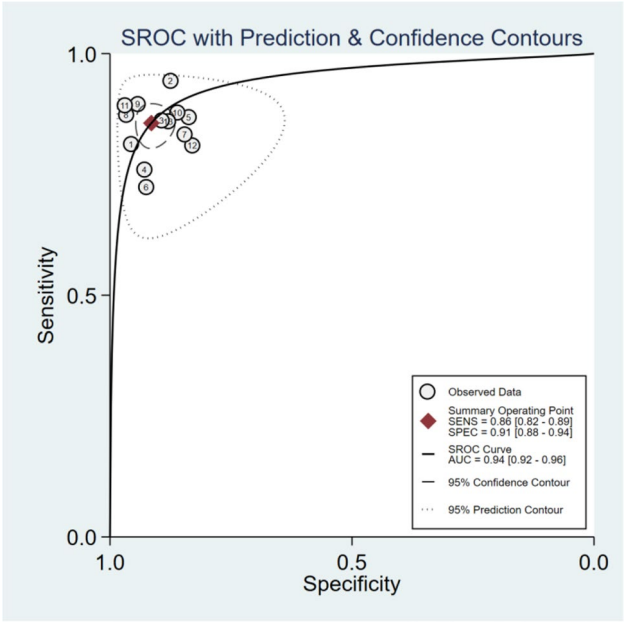


Fig. 3 Summary receiver operating characteristic curve (sROC) for predictive studies

indicating excellent overall diagnostic performance (Fig. 3). Most data points clustered in the upper-left region of the curve, reflecting generally high diagnostic accuracy across studies. The 95% confidence region in the SROC plot was relatively narrow, whereas the prediction region was comparatively wider, suggesting that although heterogeneity was substantial, the pooled estimates remained stable at the population level.

The Deeks' funnel plot (Fig. 4) revealed no obvious asymmetry in the distribution between $1/\sqrt{ESS}$ and the log diagnostic odds ratio. The regression line was approximately vertical, and scatter points were largely symmetrical, with one small-sample study located near the X-axis. The P-value of the Deeks' test was 0.63, indicating no statistically significant evidence of publication bias ($P > 0.05$). These findings suggest that no strong small-study effects or publication bias were detected.

Subgroup analysis

A subgroup analysis was performed based on image type and unit of analysis (Fig. 5). Among the 13 studies included, 8 used dental radiographs, while 5 employed clinical intraoral images. Regarding the unit of analysis,

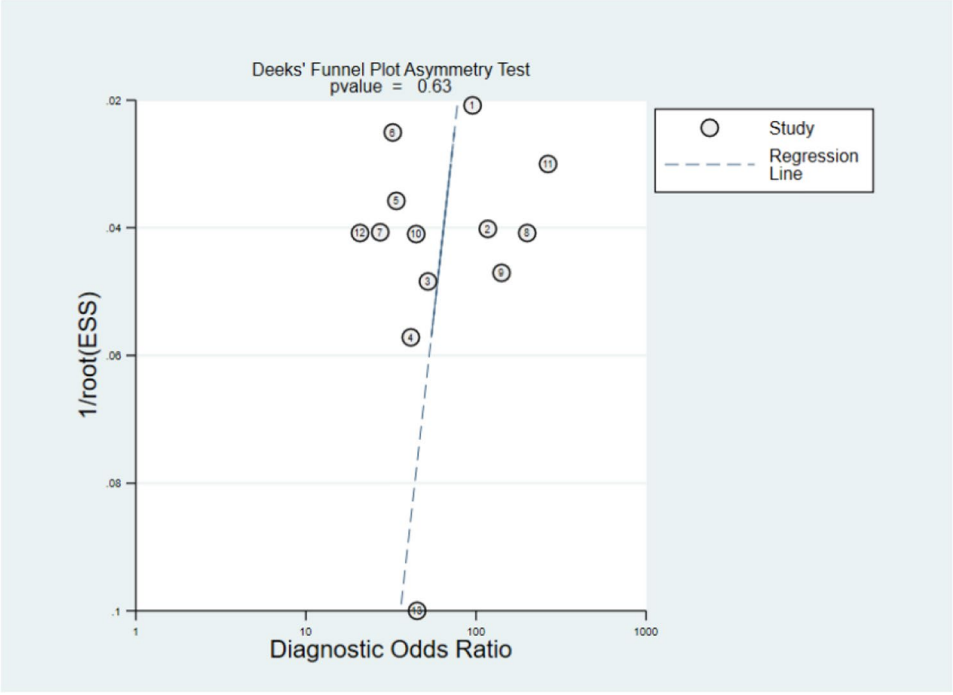


Fig. 4 Deeks' funnel plot for publication bias assessment

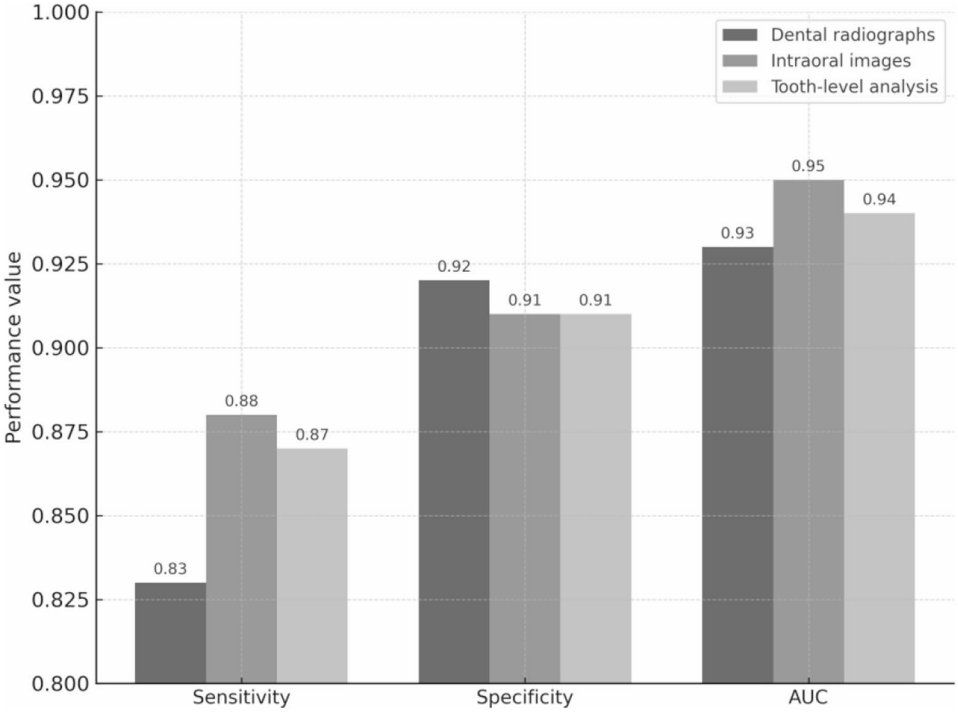


Fig. 5 Subgroup performance comparison of AI models for dental caries diagnosis

11 studies evaluated individual teeth, one assessed entire images, and one assessed individual tooth surfaces. In tooth-level studies, each individual tooth was treated as a separate diagnostic unit, with AI predictions and reference standard labels assigned per tooth rather than per image. Given the limited number of studies in the latter two categories, and considering that tooth-level analysis more closely reflects routine clinical decision-making, only studies using the tooth as the unit of analysis were included in the subgroup analysis.

Among the 8 studies using dental radiographs, the pooled sensitivity was 0.83 (95% CI: 0.79–0.87) and the pooled specificity was 0.92 (95% CI: 0.87–0.95), with an AUC of 0.93 (95% CI: 0.90–0.95). Heterogeneity analysis revealed high heterogeneity, with I^2 values of 89.30% for sensitivity and 94.70% for specificity. For the 5 studies using clinical intraoral images, the pooled sensitivity was 0.88 (95% CI: 0.83–0.92) and the pooled specificity was 0.91 (95% CI: 0.86–0.94), with an AUC of 0.95 (95% CI: 0.93–0.97). Sensitivity and specificity I^2 values were 93.39% and 96.00%, respectively, also indicating substantial heterogeneity.

In addition, the pooled results from the 11 studies using the individual tooth as the unit of analysis demonstrated a sensitivity of 0.87 (95% CI: 0.83–0.89) and a specificity of 0.91 (95% CI: 0.88–0.94), with an AUC of 0.94 (95% CI: 0.92–0.96). Heterogeneity remained high in this subgroup, with I^2 values of 88.13% for sensitivity and 95.71% for specificity.

Certainty of evidence

Using the GRADE framework, the certainty of evidence for pooled sensitivity, specificity, and AUC was rated as low to moderate. Downgrades were mainly due to high heterogeneity (I^2 > 90%), moderate methodological quality in most studies, variability in datasets and reference standards, and potential selective reporting. While results indicate promising diagnostic accuracy of AI models for dental caries, the current evidence base remains insufficient for strong clinical recommendations.

Discussion

This systematic review and meta-analysis assessed the diagnostic performance of AI models for the diagnosis of dental caries. A key strength of this study was its emphasis on the binary diagnostic accuracy of AI models in determining whether dental caries are present, rather than on broader tasks such as lesion segmentation or localization. This targeted approach narrowed the scope of the review, allowing for stricter inclusion criteria that captured only studies with well-defined diagnostic objectives and standardized performance metrics. This approach improved the homogeneity and comparability of the meta-analysis, thereby strengthening the representativeness and clinical relevance of its conclusions. Unlike previous reviews covering multiple task types [52, 53], this review focused on a core task with clear clinical translational potential, offering stronger evidence for applying AI to caries screening and diagnostic assistance.

The meta-analysis demonstrated that AI models achieved good diagnostic accuracy, with pooled sensitivity of 0.86 (95% CI: 0.82–0.89), specificity of 0.91 (95% CI: 0.88–0.94), and an AUC of 0.94 (95% CI: 0.92–0.96), indicating substantial potential for clinical application in dental caries diagnosis. Subgroup analyses further refined and validated the applicability and robustness of these findings. These findings are broadly consistent with two recent systematic reviews by Carvalho [25] and Ammar [26], which also reported high sensitivity and specificity of AI for proximal caries detection based on bitewing radiographs. However, those reviews were restricted to a single imaging modality and did not focus specifically on binary diagnostic classification or methodological reporting quality. By contrast, the present study extends this evidence base by synthesizing results across multiple imaging modalities and analytical units, while integrating diagnostic performance with CLAIM and QUADAS-2 assessments. This broader analytical framework provides a more comprehensive and clinically informative evaluation of AI-based caries detection.

The subgroup analysis further revealed the influence of image type and unit of analysis on the diagnostic performance of AI models. Models using clinical intraoral images showed higher sensitivity (0.88) and AUC (0.95), likely because high-resolution color images could clearly display changes in lesion color, surface texture, and morphology, enabling deep learning models to capture subtle pathological features [12, 54]. However, such images were more prone to interference from lighting, saliva reflections, calculus, and staining, which might account for their slightly lower specificity compared with dental radiographs. In contrast, models using dental radiographs showed a slight specificity advantage (0.92), likely because they provided a stable view of hard tissue density changes and were less influenced by lighting or color factors [52, 55]. Nevertheless, radiographs had limitations in visualizing early enamel caries, which might lead to underestimation of lesion extent and reduced sensitivity.

For the unit of analysis, tooth-level studies showed pooled sensitivity of 0.87 and specificity of 0.91, indicating balanced performance and close alignment with routine clinical workflows. Using the tooth as the analytical unit can reduce cumulative error from detecting multiple objects within a single image and minimize bias from inconsistent local annotations, in line with the "unit-of-analysis" consistency principle proposed in previous studies [56, 57]. Therefore, future AI model development and validation should prioritize tooth-level analysis. However, subgroup analyses by dentition type were limited by the small and imbalanced number of available studies and should therefore be interpreted cautiously. Combining tooth-level analysis with multimodal imaging may further enhance diagnostic precision by leveraging the complementary strengths of different modalities.

With respect to publication bias, although the Deeks' funnel plot test did not reveal statistically significant bias (P = 0.63) and the data points between $1/\sqrt{ESS}$ and log(DOR) were generally symmetrically distributed, this

result should be interpreted with caution. First, the inclusion of only 13 studies in the meta-analysis limited the statistical power for detecting bias. Second, AI research is often published with a focus on "high-performance metrics." This practice may encourage selective reporting, in which well-performing models are preferentially reported while failed experiments or negative results are omitted. Such bias may not be fully captured by the current funnel plot analysis.

Despite the overall favorable diagnostic performance observed in this meta-analysis, a first major limitation was the substantial heterogeneity across studies (I^2 > 90%), which indicates that the pooled estimates should be interpreted cautiously and may not be directly generalizable to all clinical settings. Several methodological and clinical factors may have contributed to this variability. One important source of heterogeneity was the imaging modality. Although subgroup analyses were conducted for dental radiographs and clinical intraoral photographs, heterogeneity remained high within each subgroup. This may be attributable to differences in image resolution, contrast, acquisition protocols, and the visual presentation of carious lesions across imaging systems and clinical settings.

In addition to imaging modality, substantial heterogeneity was also driven by variability in model architectures, reference standards, and dataset characteristics. The included studies employed a wide range of convolutional neural networks, transformer-based models, and self-supervised learning frameworks, each with different feature extraction mechanisms, training data requirements, and optimization strategies, which are known to influence diagnostic performance. However, the limited number of studies within each specific model category and their highly unbalanced distribution precluded formal model-type–specific subgroup analyses or meta-regression in the present study. Moreover, although most studies relied on expert-based annotations, key methodological details such as assessor blinding, the number of evaluators, and inter-observer agreement were often incompletely reported, with kappa statistics available in only 4 of 25 studies. Importantly, expert-based labeling should not be regarded as a uniform or error-free gold standard, particularly for early or proximal lesions, and such variability in reference standards introduces uncertainty into both sensitivity and specificity estimates across studies.

Dataset-level factors may have further contributed to heterogeneity. Most studies were based on single-center datasets with variable participant demographics, dentition types, and image quality. In addition, caries prevalence in the test sets ranged widely (15.68%–80.63%), reflecting spectrum effects that can substantially influence diagnostic metrics and limit the generalizability of pooled estimates to routine clinical populations.

A second major limitation of the current evidence base relates to the quality and transparency of reporting, as reflected by the CLAIM assessment. Among the 25 included studies, only one was rated as high quality, while the majority were classified as moderate quality and three were considered low quality. The primary deficiencies were concentrated in two critical domains: model interpretability and reproducibility. More than 60% of the studies did not provide visual explanations of model decision-making (e.g., Grad-CAM or LIME), which may undermine clinical interpretability, acceptability, and trustworthiness. In addition, most studies failed to release source code, training protocols, or model weights, severely limiting opportunities for independent validation and reproducibility. Only one study provided an online access link to the developed model, whereas the remaining studies disclosed no such information. It should also be acknowledged that CLAIM scoring was based exclusively on publicly available information without contacting original authors, which may have contributed to lower ratings in certain domains. Nevertheless, from the perspective of clinical translation and scientific reproducibility, such reporting gaps represent a substantive limitation of the current literature. These shortcomings restrict the transferability and adaptability of AI models across diverse clinical environments and hinder their inclusion in future large-scale comparative analyses or ensemble learning frameworks.

Taken together, the heterogeneity in diagnostic performance and the methodological limitations identified through the CLAIM assessment indicate that AI-based caries detection should be interpreted within a clearly defined clinical context. From a clinical perspective, current AI models are best viewed as complementary decision-support and screening tools rather than substitutes for professional judgment, as clinicians integrate multiple sources of information beyond image-based assessment alone. In this setting, the relatively high pooled sensitivity observed in this meta-analysis suggests that AI may help reduce missed caries cases in routine practice; however, because most models are trained and evaluated using binary labels without lesion-stage stratification, they may be more reliable for identifying established rather than very early lesions. Accordingly, the clinical utility of these models is likely to vary across populations with different caries prevalence, with potentially greater value in high-prevalence settings and an increased risk of false-positive findings in low-prevalence contexts.

For translation into routine practice, regulatory, ethical, and workflow considerations will be equally important. Regulatory approval will depend not only on diagnostic accuracy but also on evidence of robustness,

transparency, and generalizability across diverse populations and imaging systems. Ethically, the deployment of AI in dental diagnostics requires careful attention to issues of accountability, automation bias, and the preservation of professional oversight. From a workflow perspective, AI systems are most likely to be adopted successfully when they are seamlessly integrated into existing imaging and reporting platforms, providing real-time, interpretable decision support without increasing clinician burden. Addressing these aspects, together with methodological standardization and external validation, will be essential for enabling safe, effective, and scalable clinical implementation of AI-based caries detection.

Conclusions

This systematic review and meta-analysis evaluated the diagnostic performance of AI models for dental caries detection. The pooled results suggest that AI systems demonstrate promising diagnostic accuracy, with a sensitivity of 0.86, a specificity of 0.91, and an AUC of 0.94. Subgroup analyses further indicate that favorable performance is observed in studies using dental radiographs and tooth-level analysis. However, these findings should be interpreted with caution due to substantial heterogeneity, variability in reference standards, and limitations in reporting quality, as reflected by the low-to-moderate certainty of evidence. Future well-designed, multicenter studies with standardized diagnostic criteria and external validation are needed to confirm these results and support broader clinical implementation.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12903-026-07770-4.

Supplementary Material 1.

Supplementary Material 2.

Authors' contributions

Jing Lai, Shanshan Guo, and Xueman Wang conceived and designed the study and drafted the manuscript. Ke Wang, Xulan Li, and Xin Yu contributed to data collection, analysis, and interpretation, and provided critical revisions to the manuscript. All authors reviewed the findings, commented on manuscript drafts, and approved the final version for submission.

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Data availability

All data generated or analyzed in this study are included in this article and its online supplementary materials. Additional inquiries can be addressed to the corresponding author.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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